

An Analysis of Vision transformer

Vision transformer

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Vision transformer -- Part 1

Sharpening: The teacher network's output is sharpened using a softmax function with a lower temperature. This makes the teacher more "confident" in its predictions, forcing the student to learn more meaningful representations to match the teacher's sharpened output. **Centering:** The teacher network's output is centered by averaging it with its previous outputs. This prevents the teacher from becoming biased towards any particular output value, encouraging the student to learn a more diverse set of features. In January 2024, Meta AI Research released an updated version called DINOv2 with improvements in architecture, loss function, and optimization technique. It was trained on a larger and more diverse dataset. The features learned by DINOv2 were more transferable, meaning it had better performance in downstream tasks. In August 2025, Meta AI Research released DINOv3, an update to DINOv2. It introduced image-text alignment like CLIP. It scaled up the model to 7B parameters and the training dataset to 1.7B images (obtained by diversity-sampling an initial dataset with 17B images). Architecturally, it introduced two improvements: Gram anchoring and axial RoPE (Rotary Positional Embeddings) with jittering. Gram anchoring applies teacher-student self-distillation for the Gram matrix between the feature vectors of the patches of an image. It avoids the previously observed problem of degradation of dense feature maps: While performance on global tasks (like classification) continued to improve, performance on dense tasks (like segmentation) would peak early and then decline, with feature maps becoming noisy. Axial RoPE makes the model more robust to varying image resolutions, scales, and aspect ratios.

Vision transformer -- Part 2

The Masked Autoencoder took inspiration from denoising autoencoders and context encoders. It has two ViTs put end-to-end. The first one ("encoder") takes in image patches with positional encoding, and outputs vectors representing each patch. The second one (called "decoder", even though it is still an encoder-only Transformer) takes in vectors with positional encoding and outputs image patches again.

Vision transformer -- Part 3

ViT-VQGAN In ViT-VQGAN, there are two ViT encoders and a discriminator. One encodes 8x8 patches of an image into a list of

vectors, one for each patch. The vectors can only come from a discrete set of "codebook", as in vector quantization. Another encodes the quantized vectors back to image patches. The training objective attempts to make the reconstruction image (the output image) faithful to the input image. The discriminator (usually a convolutional network, but other networks are allowed) attempts to decide if an image is an original real image, or a reconstructed image by the ViT. The idea is essentially the same as vector quantized variational autoencoder (VQVAE) plus generative adversarial network (GAN). After such a ViT-VQGAN is trained, it can be used to code an arbitrary image into a list of symbols, and code an arbitrary list of symbols into an image. The list of symbols can be used to train into a standard autoregressive transformer (like GPT), for autoregressively generating an image. Further, one can take a list of caption-image pairs, convert the images into strings of symbols, and train a standard GPT-style transformer. Then at test time, one can just give an image caption, and have it autoregressively generate the image. This is the structure of Google Parti.

Vision transformer -- Part 4

Applications ViT have been used in many computer vision tasks with excellent results and in some cases even state-of-the-art, such as in image classification, object detection, video deepfake detection, image segmentation, anomaly detection, image synthesis, cluster analysis, autonomous driving. ViT had been used for image generation as backbones for GAN and for diffusion models (diffusion transformer, or DiT). DINO has been demonstrated to learn useful representations for clustering images and exploring morphological profiles on biological datasets, such as images generated with the Cell Painting assay. In 2024, a 113 billion-parameter ViT model was designed (the largest ViT to date) for weather and climate prediction, and trained on the Frontier supercomputer with a throughput of 1.6 exaFLOPs.

Vision transformer -- Part 5

LayerNorm immediately after each attention and feedforward layer ("res-post-norm"); scaled cosine attention to replace the original dot product attention; log-spaced continuous relative position bias, which allows transfer learning across different window resolutions.

Vision transformer -- Part 6

A space attention layer is a self-attention layer where each query patch $q_{x,y,t}$ attends to only the key and value patches $k_{x',y',t'}, v_{x',y',t'}$ such that $t = t'$. A

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time attention layer is where the requirement is $x' = x, y' = y$ $\{\displaystyle x'=x,y'=y\}$ instead. The TimeSformer also considered other attention layer designs, such as the "height attention layer" where the requirement is $x' = x, t' = t$ $\{\displaystyle x'=x,t'=t\}$. However, they found empirically that the best design interleaves one space attention layer and one time attention layer.

Vision transformer -- Part 7

Others Other examples include the visual transformer, CoAtNet, CvT, the data-efficient ViT (DeiT), etc. In the Transformer in Transformer architecture, each layer applies a vision Transformer layer on each image patch embedding, add back the resulting tokens to the embedding, then applies another vision Transformer layer.

Vision transformer -- Part 8

A vision transformer (ViT) is a transformer designed for computer vision. A ViT decomposes an input image into a series of patches (rather than text into tokens), serializes each patch into a vector, and maps it to a smaller dimension with a single matrix multiplication. These vector embeddings are then processed by a transformer encoder as if they were token embeddings. ViTs were designed as alternatives to convolutional neural networks (CNNs) in computer vision applications. They have different inductive biases, training stability, and data efficiency. Compared to CNNs, ViTs are less data efficient, but have higher capacity. Some of the largest modern computer vision models are ViTs, such as one with 22B parameters. Subsequent to its publication, many variants were proposed, with hybrid architectures with both features of ViTs and CNNs. ViTs have found application in image recognition, image segmentation, weather prediction, and autonomous driving.

Vision transformer -- Part 9

Further reading Zhang, Aston; Lipton, Zachary; Li, Mu; Smola, Alexander J. (2024). "11.8. Transformers for Vision". Dive into deep learning. Cambridge New York Port Melbourne New Delhi Singapore: Cambridge University Press. ISBN 978-1-009-38943-3. Steiner, Andreas; Kolesnikov, Alexander; Zhai, Xiaohua; Wightman, Ross; Uszkoreit, Jakob; Beyer, Lucas (June 18, 2021). "How to train your ViT? Data, Augmentation, and Regularization in Vision Transformers". arXiv:2106.10270 [cs.CV].